

Exercises for Section 10.2

Directions: For exercises 1 - 7, identify the vertices, foci, and equations of the asymptotes.

1. $\frac{x^2}{25} - \frac{y^2}{36} = 1$

2. $\frac{x^2}{100} - \frac{y^2}{9} = 1$

3. $\frac{y^2}{4} - \frac{x^2}{81} = 1$

4. $9y^2 - 4x^2 = 1$

5. $\frac{(x-1)^2}{9} - \frac{(y-2)^2}{16} = 1$

6. $\frac{(y-6)^2}{36} - \frac{(x+1)^2}{16} = 1$

7. $\frac{(x-2)^2}{49} - \frac{(y+7)^2}{49} = 1$

Directions: For exercises 8 - 10, find the equations of the asymptotes for each hyperbola.

8. $\frac{y^2}{9} - \frac{x^2}{9} = 1$

9. $\frac{(x-3)^2}{25} - \frac{(y+4)^2}{4} = 1$

10. $\frac{(y-3)^2}{9} - \frac{(x+5)^2}{36} = 1$

Directions: For exercises 11 - 16, sketch a graph of the hyperbola, labeling vertices and foci.

11. $\frac{x^2}{49} - \frac{y^2}{16} = 1$

12. $\frac{x^2}{64} - \frac{y^2}{4} = 1$

13. $\frac{y^2}{9} - \frac{x^2}{25} = 1$

14. $81x^2 - 9y^2 = 1$

15. $\frac{(y+5)^2}{9} - \frac{(x-4)^2}{25} = 1$

16. $\frac{(x-2)^2}{8} - \frac{(y+3)^2}{27} = 1$

Directions: For exercises 17 - 22, given information about the graph of the hyperbola, find its equation.

17. Vertices at (3,0) and (-3,0) and one focus at (5,0).

18. Vertices at (0,6) and (0,-6) and one focus at (0,-8).

19. Vertices at (1,1) and (11,1) and one focus at (12,1).

20. Center: (0,0); vertex: (0,-13); one focus: $(0, \sqrt{313})$.

21. Center: (4,2); vertex: (9,2); one focus: $(4 + \sqrt{26}, 2)$.

22. Center: (3,5); vertex: (3,11); one focus: $(3, 5 + 2\sqrt{10})$.